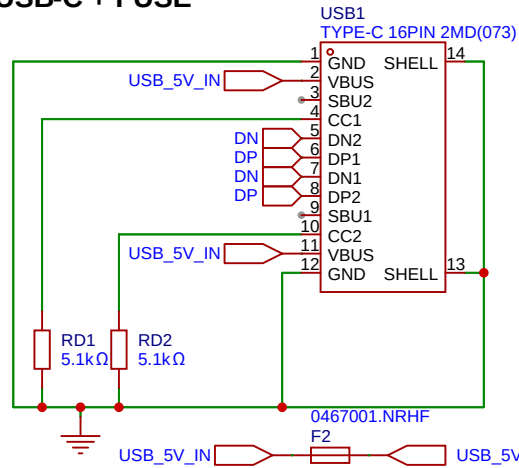
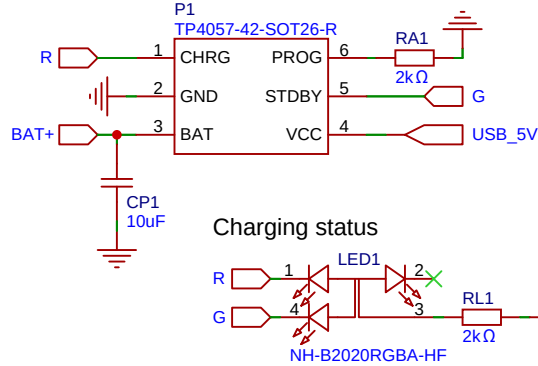


USB-C + FUSE



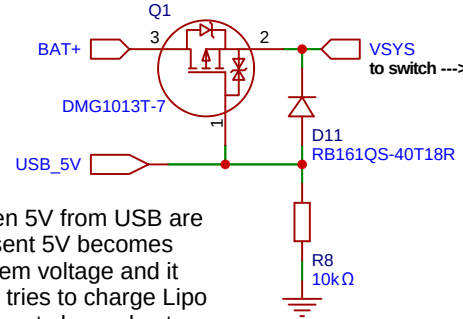
LiPo charger / LED status

Charging current is 400 mA. 500 mA is max for this chip



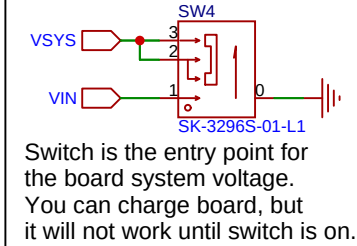
Power Path

Transistor can handle 460 mA, average pow. cons. of the board is 0.4W, so should be more than fine

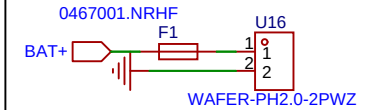


When 5V from USB are present 5V becomes system voltage and it also tries to charge Lipo if it's not charged yet.

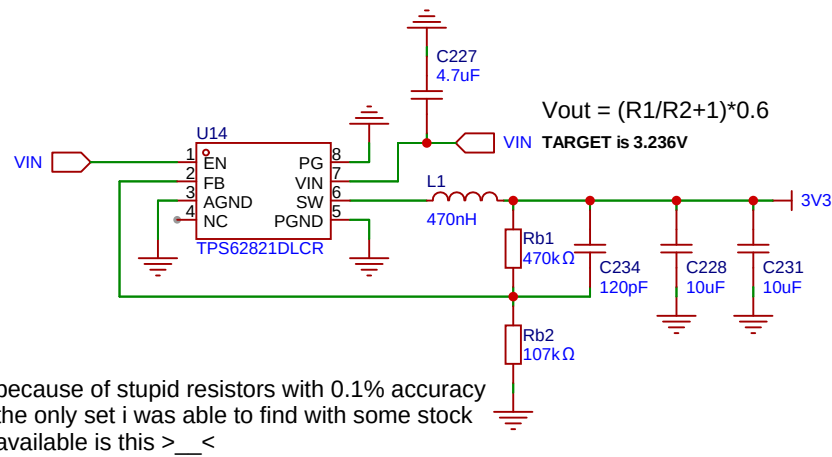
Switch (system voltage)



Battery Connector + FUSE

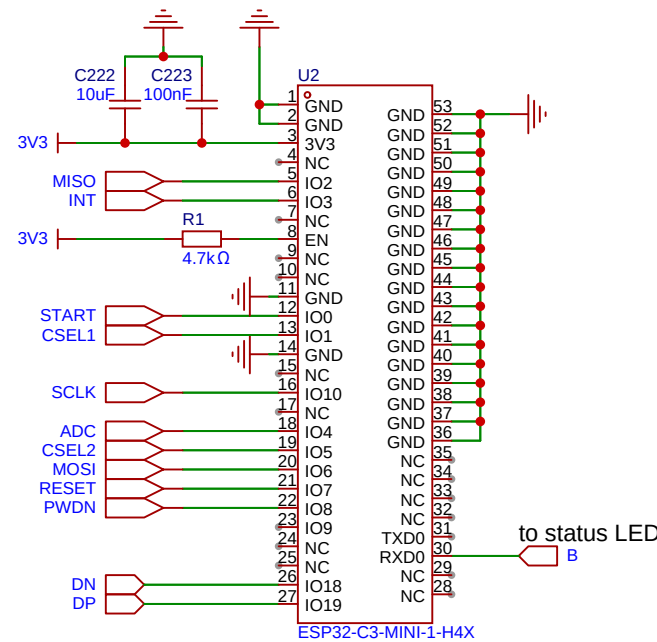


3.3V Buck converter --- TO ESP AND ADCs digital part



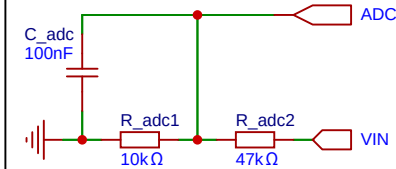
because of stupid resistors with 0.1% accuracy the only set i was able to find with some stock available is this >__<

ESP32 C3 MINI



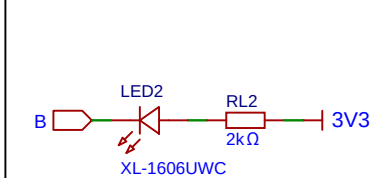
System Voltage Sens

You will see pure battery voltage or charging voltage if usb 5V are present



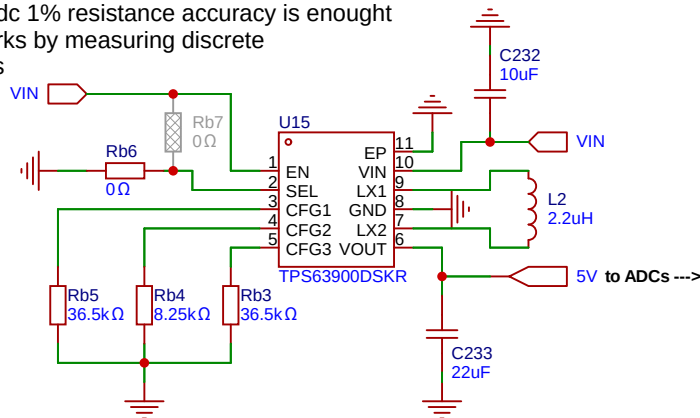
Cap is needed to help to smooth out voltage noise

ESP status LED



5V Buck-Boost converter --- TO ADCs

For this dcdc 1% resistance accuracy is enough since it works by measuring discrete resistances



Schematic	BCI_ADS1299			Update Date	2025-05-24
				Create Date	
Page	ESP and Power			Part Number	
Drawn		BCI_nikki_corp_ver_1.0			
Reviewed					
		VER	SIZE	PAGE	1 OF 3
		V3.0	A4	Designed by --nikki--	

RANDOM KNOWLEDGE

Hi Mojtaba,

I'll try to be more clear.

Q: in total what does the BIASIN pin do and whats it's role?

A: In the event that the BIAS electrode becomes disconnected from the patient, by having BIASOUT connected to BIASIN, the BIASOUT signal can be routed through BIASIN and back to the patient through one of the input electrodes. This way, the patient stays biased and the measurements stay within the common-mode input range of the PGA. Of course, one measurement will be "lost" since it's being used to bias the patient.

Q: it should be done in both cascade or daisy chain mode or it's just for daisy chain?

A: it's the same for both modes.

BIASOUT and BIASIN are shorted, because BIASIN is going towards mux right away. So as i understood if BIASIN is off MUX will not use BIAS voltage to shift signal from 0 to as i undestood positive AVDD/2. For unipolar powering AVSS is equal to 0 and it means it's just GROUND. Check 9.2 Functional Block Diagram for BIASIN

FOR UNIPOLAR POWERING SCHEME AVSS IS 0 - BASICALLY GND AS WELL AS DIGITAL GROUND

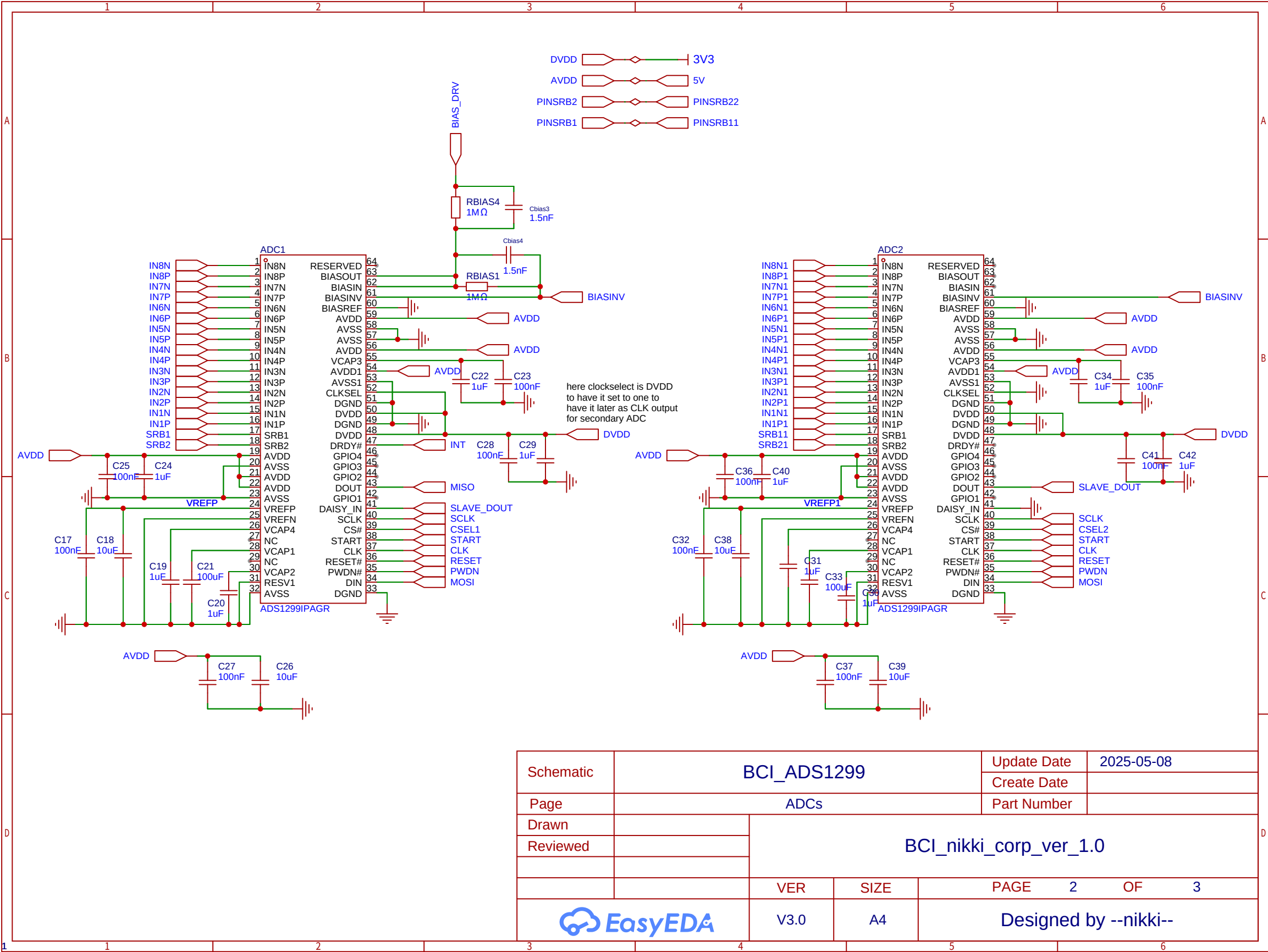
It's important to keep decoupling capacitors for all power pins. You can combine several pins ino one decoupling capacitor output if they are supa close to eachother. But in general - Datasheet says to have decoupling cap for all of them - 1uF and 0.1uF

This is a very interesting problem! Let me first explain what the SRB pins do and what the BIAS pin does. The SRB pins are essentially inputs for a reference input that you intend to connect to multiple channels. The SRB2 pin can be connected to individual channels' positive inputs. The SRB1 pin can be connected to all channels' negative inputs.

The BIAS pin is the equivalent of a right-leg driver for an ECG measurement. It will output a signal that attempts to keep the patient's common-mode voltage within the valid range for the ADC.

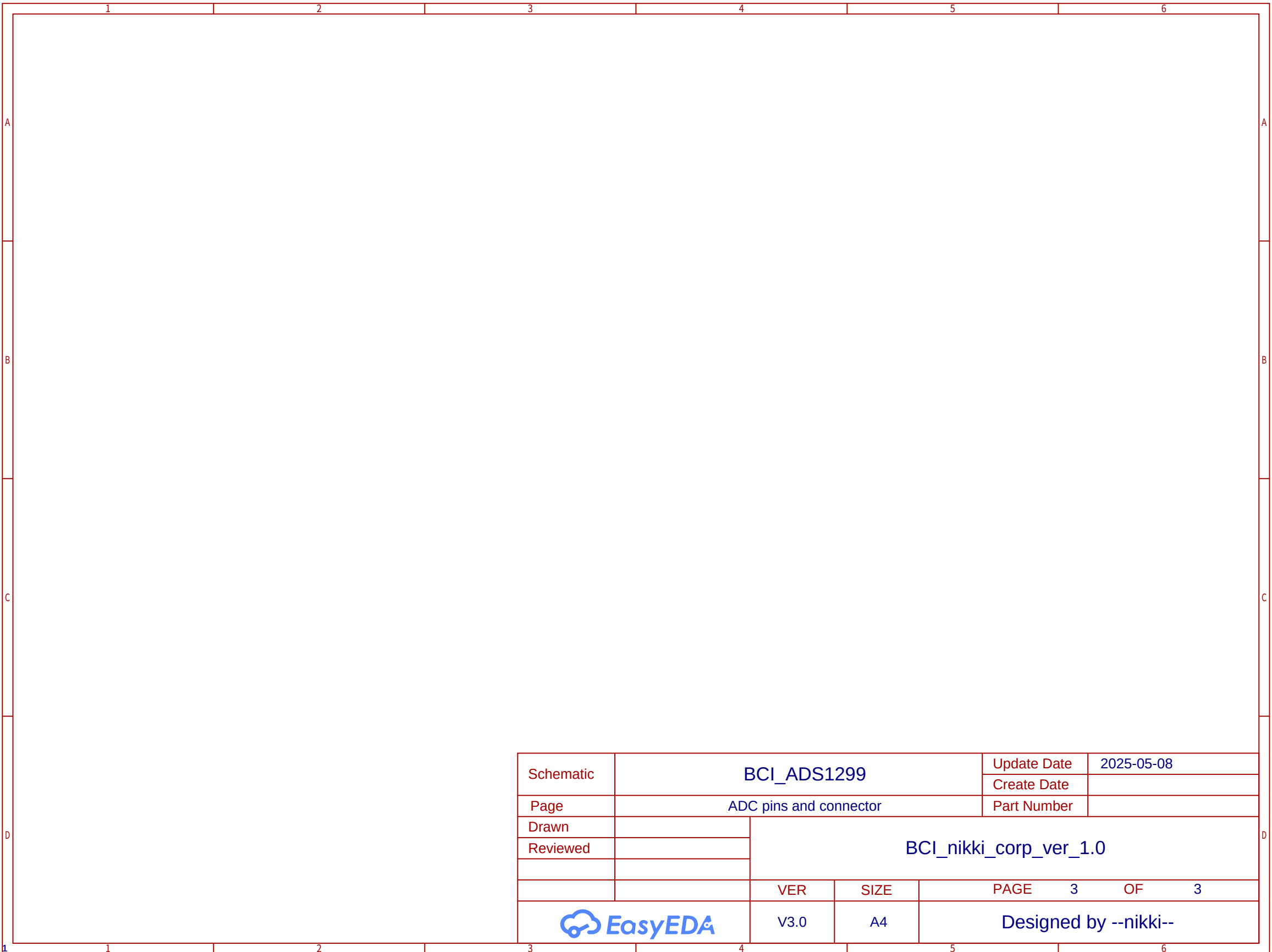
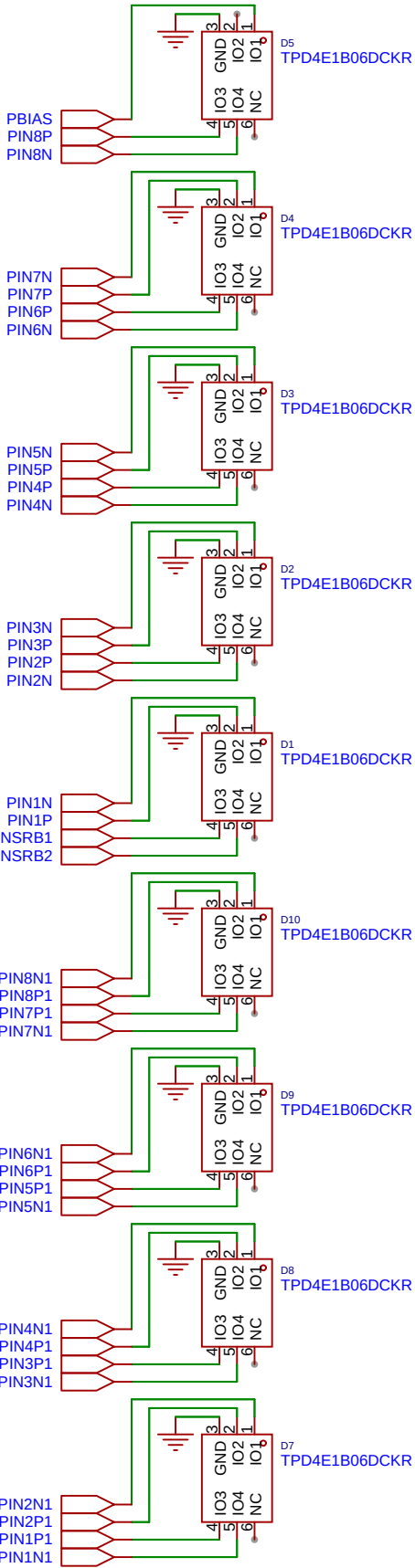
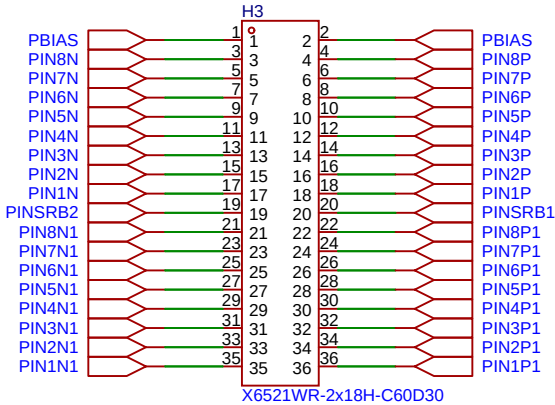
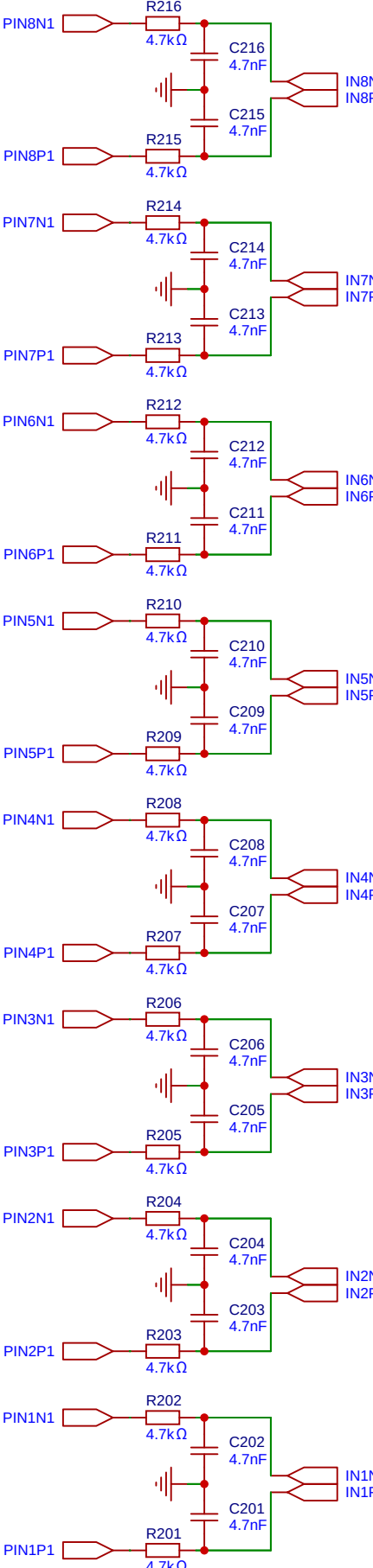
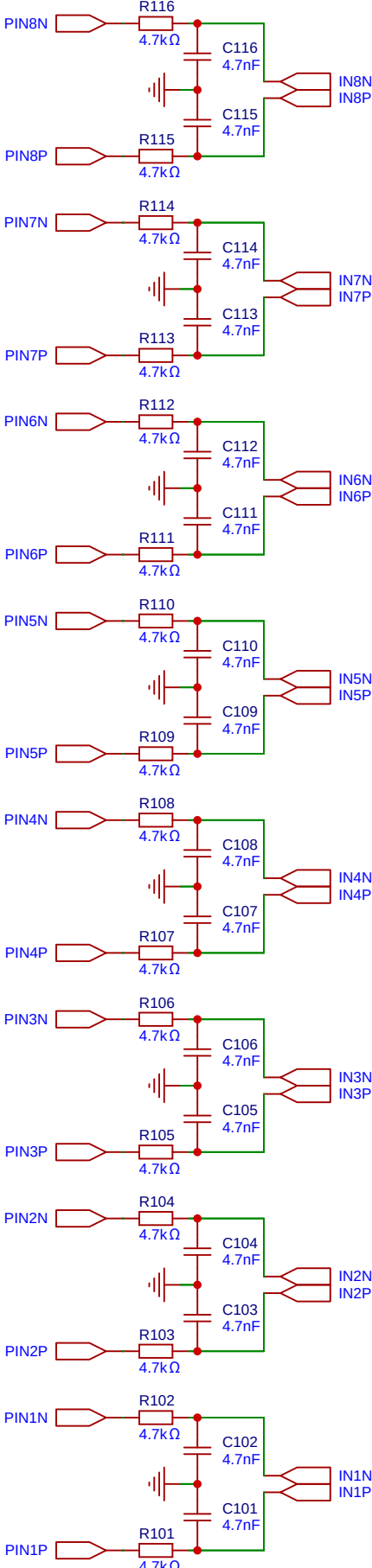
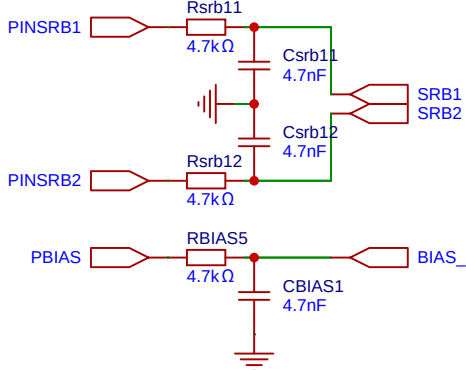
This datasheet was actually just updated with a helpful section that discusses the various functions of these pins. You can find it here starting on page 65.

In your case, though, since it is intended to be so modular, I recommend you simply connect all INxN inputs to analog ground (assuming you are using bipolar supplies) and find each electrode voltage in a single-ended fashion. Then you can average each sample to calculate a "reference voltage" to then analyze each individual channel.



Schematic	BCI_ADS1299			Update Date	2025-05-08
				Create Date	
Page	ADCs			Part Number	
Drawn		BCI_nikki_corp_ver_1.0			
Reviewed					
		VER	SIZE	PAGE	2 OF 3
EasyEDA		V3.0	A4	Designed by --nikki--	

4.7 kOhm and 4.7nF are from TI evaluaion board datasheet.
They provide beter filtering of high frequency than having
lower resistance and capacitance.



Schematic	BCI_ADS1299			Update Date	2025-05-08	
				Create Date		
Page	ADC pins and connector			Part Number		
Drawn	BCI_nikki_corp_ver_1.0					
Reviewed						
		VER	SIZE	PAGE	3	OF 3
		V3.0	A4	Designed by --nikki--		
3		4		5		6